

S-TTRL: SAMPLE-WISE TEST-TIME REINFORCEMENT LEARNING FOR LARGE LANGUAGE MODELS

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ABSTRACT

Test-time adaptation has emerged as a compelling way to improve large language models (LLMs) on out-of-distribution queries, yet most Test-Time Reinforcement Learning (TTRL) methods assume *batch-wise* access to the test set during inference—an assumption misaligned with real deployments that are interactive, privacy-sensitive, or streaming. We introduce *Sample-wise Test-Time Reinforcement Learning (S-TTRL)*¹, a per-query adaptation framework that improves the model using only the single unlabeled prompt at hand. S-TTRL addresses the unique challenges of single-sample adaptation—signal sparsity, overfitting/forgetting, and the stability–latency trade-off—by (i) generating multiple candidates and aggregating them into a robust reference (e.g., voting/consistency), (ii) deriving rewards from verifiable or internal signals, and (iii) performing a brief policy-optimization step before emitting the final answer. Across diverse math and reasoning benchmarks (e.g., AIME 2024, AMC, MATH-500, GPQA), S-TTRL outperforms other sample-wise baselines like SLOT and MEMO. However, our findings also reveal that sample-wise adaptation does not match the performance of online TTRL, highlighting an inherent trade-off between per-sample flexibility and performance stability.

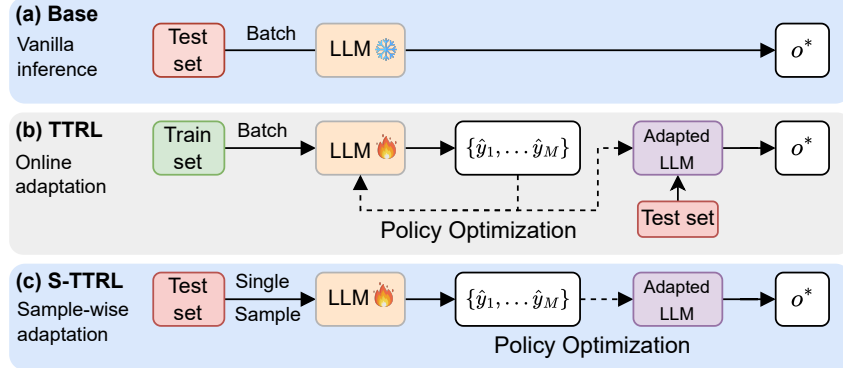


Figure 1: **From batch-wise to sample-wise test-time adaptation.** (a) *Standard inference*: the frozen LLM answers each test query without adaptation. (b) *Batch-wise TTRL* (Zuo et al., 2025): the model performs online policy optimization on the train set, then infers with the adapted parameters on the test set. (c) *S-TTRL (ours)*: adaptation is performed *per single query*, using only that query’s unlabeled prompt and internally/verifiably derived rewards, followed by the final prediction for the same query.

1 INTRODUCTION

Large language models (LLMs) have recently benefited from reinforcement learning (RL) to strengthen long chain-of-thought (CoT) reasoning, as illustrated by DeepSeek-R1 (Guo et al., 2025) and Qwen2.5-7B (Yang et al., 2025), building on the broader success of RL for reasoning (Wei

¹Code available at: <https://anonymous.4open.science/r/sttrl-8E23/README.md>

et al., 2022). Beyond verifiable rewards (RLVR), Reinforcement Learning from Internal Feedback (RLIF) (Zhao et al., 2025) replaces external supervision with intrinsic model signals—e.g., using confidence as reward in GRPO (Shao et al., 2024).

A parallel line of work adapts models at *test time*. Test-Time Adaptation/Training (TTA/TTT) (Xiao & Snoek, 2024) exploits unlabeled test data to reduce train–test mismatch; TTT often includes a preparatory adaptation stage (Yuan et al., 2024). While originally developed in vision for robustness under distribution shift and corruption (Sun et al., 2020a; Zhang et al., 2022), these ideas have recently been transferred to LLM reasoning. Notably, Test-Time Reinforcement Learning (TTRL) (Zuo et al., 2025) applies RL-based updates to improve LLM performance on math and reasoning benchmarks, and concurrent work explores verifier-guided continuous test-time training for LLMs (Moradi et al., 2025). Figure 1 contrasts standard inference, batch-wise TTRL, and our sample-wise variant.

Despite their promise, existing TTRL methods typically operate in a *batch-wise online* manner (Xiao & Snoek, 2024): they assume access to a sizeable train set on which they optimize, and a complete test batch during inference. This assumption is misaligned with many deployment scenarios—streaming queries, privacy or memory constraints, and non-stationary test streams—where large batches are unavailable or undesirable. Moreover, batch-wise optimization can entangle samples from different sub-distributions and may induce instability when the test stream drifts over time.

To align test-time learning with real-world usage, we propose *Sample-wise Test-Time Reinforcement Learning (S-TTRL)*, which adapts the model *per query* using only that query’s unlabeled prompt (Fig. 1c). While more practical, the setting introduces distinct challenges: (i) *signal sparsity/noise*—a single example yields weaker and potentially noisy feedback compared to a batch; (ii) *overfitting and forgetting*—aggressive per-sample updates can harm general capability; and (iii) *stability–latency trade-off*—updates must be effective yet lightweight to keep inference responsive.

We propose **S-TTRL** (Fig. 1c), a per-query test-time RL procedure that makes *single-sample* updates both practical and stable. Given one unlabeled prompt, S-TTRL (i) samples multiple candidates and uses a self-consistency/voting aggregator to form a robust reference, (ii) constructs a *dual-signal* reward that blends verifiable feedback with intrinsic signals (agreement with the reference, calibrated confidence), following the spirit of RLVR/RLIF (Zhao et al., 2025; Shao et al., 2024), and (iii) applies a brief GRPO-style, KL-regularized policy update in a parameter-efficient subspace (e.g., LoRA/prefix). A lightweight stability toolkit—trust-region control, variance-aware step sizing, revert-on-regret checks, and an EMA anchor to the frozen policy—mitigates overfitting and forgetting while keeping latency low. The adapted policy then emits the final answer for the same query. Our contributions are threefold:

- **Single-sample TTRL formulation with safe updates.** We cast per-query adaptation as a KL-regularized, parameter-efficient policy update, removing the need for batch-wise test access while preserving the benefits of TTRL-style optimization.
- **Dual-signal reward shaping.** We combine verifiable feedback with intrinsic consistency/confidence signals to improve reward signal quality in the single-sample regime, consistent with RLVR/RLIF (Zhao et al., 2025; Shao et al., 2024).
- **Stability and efficiency for streaming use.** Trust-region controls, variance-aware step sizing, revert-on-regret, and EMA anchoring deliver stable gains with modest overhead, enabling practical on-device or streaming deployment. We evaluate across math and reasoning benchmarks and ablate each component.

2 RELATED WORK

2.1 REINFORCEMENT LEARNING FOR LARGE LANGUAGE MODELS.

Recent studies have demonstrated that reinforcement learning (RL) can substantially enhance the reasoning capabilities of large language models (LLMs). Reinforcement Learning from Human Feedback (RLHF) (Christiano et al., 2017) was first introduced to align model behavior with human preferences, and later work extended its use to promote structured reasoning and self-verification. Large-scale efforts such as DeepSeek-R1 (Guo et al., 2025) have shown that reinforcement optimization at scale can further improve chain-of-thought (CoT) reasoning and problem-solving proficiency. Beyond human feedback, recent approaches have explored verifiable or intrinsic reward signals to

reduce reliance on external supervision. Reinforcement Learning with Verifiable Rewards (RLVR) (Shao et al., 2025) introduces rule-based or programmatic correctness checks to provide more reliable feedback signals, improving both training stability and objective grounding.

2.2 TEST-TIME ADAPTATION AND TRAINING.

Test-time adaptation (TTA) and test-time training (TTT) aim to mitigate domain shift between training and deployment (Sun et al., 2020b; Wang et al., 2020; Niu et al., 2022; Xiao & Snoek, 2024). In computer vision, TTA methods such as Tent (Wang et al., 2020) and EATA (Niu et al., 2022) adapt model parameters during inference by minimizing prediction entropy on unlabeled test samples. TTT (Sun et al., 2020b) extends this principle by incorporating an auxiliary self-supervised objective during training, enabling models to learn domain-invariant representations and better generalize to unseen test conditions.

2.3 TEST-TIME ADAPTATION FOR LARGE LANGUAGE MODELS

While most TTA and TTT methods were initially developed for computer vision, recent work has begun adapting these ideas to language models and reasoning tasks. Hu et al. (2025) introduce SLOT, a sample-specific optimization approach that refines the model’s hidden state before decoding, enabling per-query personalization without changing the model weights. Zuo et al. (2025) extends TTT to reinforcement learning with their approach, TTRL, where grouped rewards over multiple samples guide an online adaptation approach.

3 PRELIMINARIES

We consider an LLM policy π_θ queried at test time by a single unlabeled prompt q . The model samples G candidate outputs $\{o_i\}_{i=1}^G$. Test-time adaptation follows Group Relative Policy Optimization (GRPO) (Shao et al., 2024), where the objective aggregates per-token advantages computed from a *grouped* reward over the G candidates:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min(r_{i,t} \hat{A}_{i,t}, \text{clip}(r_{i,t}, 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t}) \right], \quad (1)$$

with $r_{i,t} = \pi_\theta(o_{i,t} | q, o_{i,<t}) / \pi_{\text{old}}(o_{i,t} | q, o_{i,<t})$ and $\hat{A}_{i,t}$ obtained by normalizing the candidate-level reward within the group. In line with Zhao et al. (2025) and our ablations, we omit an explicit KL penalty, which adds overhead without consistent gains in this setting.

Batch-wise TTRL (baseline). Test-Time Reinforcement Learning (TTRL) (Zuo et al., 2025) instantiates an *online, batch-wise* adaptation regime: at inference step t , a batch $B_t = \{q_j\}_{j=1}^m$ of test prompts is available. For each q_j , the policy samples G candidates and computes a grouped reward (e.g., via majority-vote pseudo-labeling), yielding a per-prompt GRPO objective $\mathcal{J}_{\text{GRPO}}^{(q_j)}(\theta)$ as in Eq. (1). The batch objective aggregates these per-prompt terms,

$$\mathcal{J}_{\text{TTRL}}(\theta; B_t) = \frac{1}{m} \sum_{j=1}^m \mathcal{J}_{\text{GRPO}}^{(q_j)}(\theta),$$

and parameters are updated online, $\theta_{t+1} \leftarrow \theta_t + \eta \nabla_\theta \mathcal{J}_{\text{TTRL}}(\theta_t; B_t)$, then reused for the next incoming batch B_{t+1} . This procedure assumes access to a sizeable test batch at each step and accumulates adaptation over time (the “online inference” setting (Xiao & Snoek, 2024)). In practice, rewards in (Zuo et al., 2025) are often instantiated by majority-vote correctness over the G candidates, with group-normalized advantages; as in our setting, an explicit KL penalty is typically unnecessary and can be omitted without loss (Zhao et al., 2025).

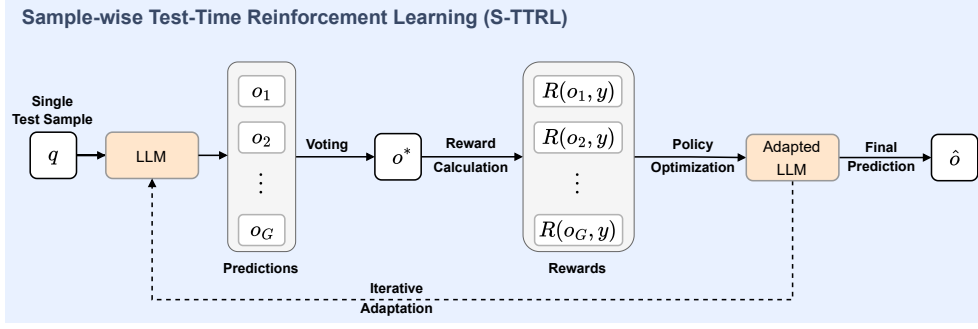


Figure 2: **S-TTRL pipeline.** For a single test query q , the frozen LLM samples G candidates $\{o_i\}$; an aggregator (e.g., majority vote/consistency) yields a reference o^* . The candidates are scored against o^* and a short GRPO update is applied on this single sample. The adapted model then emits the final prediction $\hat{o}$ for the same query.

4 METHODS

4.1 PER-QUERY WORKFLOW AND AGGREGATION

Figure 2 illustrates our proposed method, S-TTRL. Given one unlabeled prompt q , the model samples G candidate solutions $\{o_i\}_{i=1}^G$ from the frozen policy π_{θ_0} using non-greedy decoding (temperature and nucleus filtering). We remove near-duplicates by light surface normalization. Task-aware parsers extract the decision token(s): the final numeric/boxed value for AIME/AMC/MATH-500, the option letter for GPQA, and a pass/fail flag when lightweight unit tests are available.

A pseudo-label o^* is estimated from the parsed answers by majority vote. Ties are resolved by the candidate with the highest length-normalized log-likelihood of its *answer tokens* under π_{θ_0} . For numeric outputs we regard two answers as equivalent if they match within a small absolute/relative tolerance; for genuinely free-form outputs we cluster candidates by token-level edit distance and select the medoid as o^* (this reduces to voting when clusters collapse).

4.2 REWARD AND SINGLE-SAMPLE UPDATE

We use a simple rule-based reward derived solely from o^* . For discrete/numeric tasks,

$$R(o_i, q) = \mathbb{I}\{\text{match}(o_i, o^*)\},$$

where `match` respects the numeric tolerance above. For code-style prompts the reward is 1 if tests pass and 0 otherwise. No external labels or verifier models are used.

Adaptation is performed *once* on this single query via GRPO (§3): group-normalized advantages are computed from $\{R(o_i)\}_{i=1}^G$ and used to update the policy. We apply a small number of gradient steps with AdamW and global gradient-norm clipping. The update can target either the full set of model weights or a fixed subset of layers (e.g., the last few decoder blocks); the rest of the network remains frozen. Consistent with our ablations, we do not include an explicit KL penalty.

4.3 STABILITY AND DEPLOYMENT

To avoid cross-sample carryover, we reset to the frozen parameters θ_0 at the start of every query; optimizer state is not reused across queries. This stateless, per-query update—short decoding for G candidates, one brief GRPO optimization, and final decoding for the same q —keeps latency predictable. The computational cost scales with the number of sampled candidates and the small number of update steps, and does not require any batch-wise access to the test set.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Models. To assess the generalizability of S-TTRL across diverse backbone architectures, we perform experiments utilizing both base and instruct models at multiple scales. The utilized models are:

- **Math Base Models:** Qwen2.5-Math-1.5B, Qwen2.5-Math-7B (Yang et al., 2024).
- **Vanilla Base Models:** Qwen2.5-7B (Yang et al., 2025).
- **Instruct Models:** LLaMA-3.1-8B-Instruct (Grattafiori et al., 2024)

Datasets. Similar to Zuo et al. (2025), we evaluate on 3 mathematical reasoning benchmarks: AIME 2024 (Li et al., 2024), AMC (Li et al., 2024), and MATH-500 (Hendrycks et al., 2021), and on GPQA-Diamond (Rein et al., 2024): a graduate-level proof question-answering benchmark.

Evaluation. Similar to Zuo et al. (2025), we adopt the pass@1 metric to evaluate the backbone models. Following their methodology, we generate 16 responses for each question and report the average score across all responses and questions in the benchmark.

Baselines. To comprehensively assess the performance of our proposed method, S-TTRL, we compare it against two representative sample-wise test-time adaptation (TTA) methods, SLOT (Hu et al., 2025) and MEMO (Zhang et al., 2022). In addition, we include TTRL as an online adaptation baseline, representing the upper bound of adaptation performance.

- Sample-specific Language Model Optimization at Test Time (SLOT) adapts the LLM by injecting a prompt-specific vector into its final hidden layer, which is iteratively updated prior to generating the first response token.
- Marginal Entropy Minimization with One Test Point (MEMO) was originally proposed as a test-time robustness method for image classification, where a model is adapted to handle visual corruptions in the input. MEMO adapts the model on a per-sample basis by minimizing the marginal entropy of predictions obtained from multiple augmented versions of the same input. To adapt this approach for the tested benchmarks, we modify MEMO by replacing its original augmix augmentation function with paraphrases of the input generated by the LLM.

Hyperparameters. S-TTRL is trained using the Adam optimizer (learning rate 5×10^{-7} , no weight decay). A total of 64 rollouts are generated, from which the top 32 highest-reward responses are selected for GRPO. Training is performed for a single optimization epoch. For TTRL and SLOT, we use all original hyperparameters. Following Zhang et al. (2022), MEMO is trained with 16 augmentations across all experiments.

5.2 MAIN RESULTS

Table 1 reports the main results. S-TTRL consistently surpasses the sample-wise baselines (SLOT and MEMO), while online TTRL serves as an upper bound. On AIME with the 1.5B math model, S-TTRL yields a 33.3% gain, exceeding the 28.0% improvement of the best baseline. By contrast, on MATH-500 the sample-wise methods do not reliably outperform the Base model. On GPQA (Qwen2.5-Math-1.5B), none of the adaptation methods exceed Base, in contrast to Zuo et al. (2025) who reported a +4.8% improvement. Finally, for the generalist Qwen2.5-7B backbone, sample-wise methods do not improve AIME/AMC; only the online TTRL setting achieves gains. Taken together, these patterns suggest that per-query majority-vote pseudo-labels work best on short-answer math (e.g., AIME/AMC), where aggregation is reliable and the answer space is low-entropy. In long-form reasoning and knowledge-heavy settings (e.g., MATH-500/GPQA), single-sample rewards are sparser/noisier, making improvements less consistent; batching in TTRL mitigates this by stabilizing both rewards and updates.

5.3 AGGREGATION RULES FOR PSEUDO-LABELING

We ablate three aggregation rules—majority vote, self-confidence, and entropy minimization—for constructing the per-query pseudo-label o^* used by S-TTRL; results are shown in Figure 3. Across

Table 1: Pass@1 score for S-TTRL and our replicated results of TTRL (Zuo et al., 2025), SLOT (Hu et al., 2025), and MEMO (Zhang et al., 2022) compared to no adaptation. Results marked with (*) are from the original research. **Bold** values indicate the best performance among sample-wise methods, while underlined values also include the Base performance.

Base Model	Method	AIME 2024	AMC	MATH-500	GPQA	Avg.
<i>Math Base Models</i>						
Qwen2.5-Math-1.5B	Base	7.5	20.3	<u>52.0</u>	<u>32.4</u>	28.1
	<i>Online Adaptation</i>					
	+ TTRL	12.3	43.7	72.1	28.0	39.0
	<i>Sample-wise Adaptation</i>					
	+ SLOT	9.6	29.6	51.5	25.2	29.0
	+ MEMO	9.6	30.7	51.4	25.0	29.2
	+ S-TTRL	<u>10.2</u>	<u>33.2</u>	<u>51.9</u>	<u>27.5</u>	<u>30.7</u>
Qwen2.5-Math-7B	Base	12.1	13.8	<u>55.3</u>	25.1	26.6
	<i>Online Adaptation</i>					
	+ TTRL	39.6	50.4	83.4*	28.8	50.6
	<i>Sample-wise Adaptation</i>					
	+ SLOT	15.2	40.7	53.9	25.3	33.8
	+ MEMO	15.8	41.4	55.2	25.4	34.5
	+ S-TTRL	<u>16.9</u>	<u>42.2</u>	54.0	<u>25.4</u>	<u>34.6</u>
<i>Generalist Base Models</i>						
Qwen2.5-7B	Base	<u>8.5</u>	<u>32.9</u>	52.5	24.0	29.5
	<i>Online Adaptation</i>					
	+ TTRL	23.3	49.5	80.5*	30.3	45.9
	<i>Sample-wise Adaptation</i>					
	+ SLOT	3.3	21.3	39.4	28.5	23.1
	+ MEMO	4.2	21.3	53.0	29.4	27.0
	+ S-TTRL	<u>8.3</u>	<u>30.6</u>	52.3	29.3	<u>30.1</u>
<i>Instruction-Tuned Models</i>						
LLaMA3.1-8B	Base	5.0	20.7	—	—	12.9
	<i>Online Adaptation</i>					
	+ TTRL	16.7	35.5	—	—	26.1
	<i>Sample-wise Adaptation</i>					
	+ SLOT	6.7	20.3	—	—	13.5
	+ MEMO	5.6	21.2	—	—	13.4
	+ S-TTRL	<u>7.5</u>	<u>21.2</u>	—	—	<u>14.4</u>

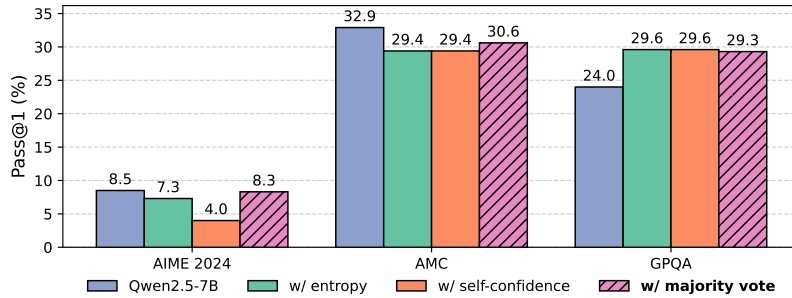


Figure 3: **Effect of aggregation rules on S-TTRL (Pass@1).** Majority vote provides the most reliable pseudo-label across backbones; confidence- and entropy-based rules do not improve performance in the single-sample regime.

the reported cells, majority voting consistently matches or exceeds self-confidence (Zhao et al., 2025) and entropy-based selection. This is consistent with the single-sample setting: confidence scores are often miscalibrated at the answer token level, and entropy minimization can favor over-confident but incorrect continuations, whereas voting reduces variance without additional calibration. We therefore use majority vote as the default pseudo-label in all main experiments and retain the alternatives as ablations.

5.4 CONCLUSION & FUTURE WORK

In this work, we introduced **Sample-wise test time reinforcement learning (S-TTRL)**, a framework that extends Reinforcement Learning with Verifiable Rewards (RLVR) to test-time adaptation (TTA) at the level of individual samples. Unlike traditional online adaptation approaches such as TTRL, which rely on access to the entire test set, S-TTRL enables lightweight, per-sample adaptation, making it more practical for deployment in dynamic or resource-constrained environments. Through extensive experiments across multiple benchmarks and model architectures, we demonstrate that S-TTRL achieves consistent improvement over existing sample-wise methods like SLOT and MEMO.

Despite the gains over other sample-wise methods, our results expose two systematic limitations. First, majority-vote pseudo-labels provide a coarse, often miscalibrated reward on long-form reasoning and knowledge-heavy tasks (e.g., MATH-500, GPQA), which makes single-sample updates high-variance and sometimes non-beneficial relative to the Base model. Second, per-query optimization cannot exploit cross-sample structure, leaving a persistent gap to online, batch-wise adaptation (TTRL) that averages signal across examples.

As future work, we plan to replace binary voting rewards with a richer *edit-and-verify* loop inspired by SEAL: the model will propose “self-edits” (synthetic training snippets and update directives) conditioned on the current query and its candidates, and we will execute a tiny finetuning step validated by downstream checks before emitting the final answer. This SEAL-style mechanism is designed to (i) densify the reward beyond a single majority decision, (ii) amortize variance by carrying forward a short-horizon buffer of vetted self-edits for similar queries, and (iii) reduce forgetting via explicit rollback criteria. We expect these changes to improve stability of sample-wise adaptation while narrowing the gap to online TTRL.

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